where e is a $k-1$ dimensional vector of ones, and we've used the identities: $\det(AB) = \det(A)\det(B)$, $\det(\text{diag}(x)) = \prod_i x_i$, and $\det(I + uv^T) = 1 + u^T v$.

We can then plug into the change of variables formula (Eq. 21) using the density of the centered Gumbel (Eq.15), the inverse of h (Eq. 22) and its Jacobian determinant (Eq. 26):

$$p(y_1, .., y_k) = \Gamma(k) \left(\prod_{i=1}^{k} \exp\left(x_i\right) \frac{y_k^\tau}{y_i^\tau} \right) \left(\sum_{i=1}^{k} \exp\left(x_i\right) \frac{y_k^\tau}{y_i^\tau} \right)^{-k} \tau^{k-1} \prod_{i=1}^{k} y_i^{-1} \tag{27}$$

$$= \Gamma(k) \tau^{k-1} \left(\sum_{i=1}^{k} \exp\left(x_i\right) / y_i^\tau \right)^{-k} \prod_{i=1}^{k} \left(\exp\left(x_i\right) / y_i^{\tau+1} \right) \tag{28}$$